⚙️ CHEMISTRY CH.2 — METALS, NON-METALS AND ALLOYS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this chapter:

1. Metals: Properties & General Characteristics

- Metals ke outermost shell mein 1–3 valence electrons hote hain; ye electrons donate karte hain aur positive ions (cations) banate hain. *[Q143, Q145, Q147, Q229]*
- Metals generally solid hote hain room temperature par (except Mercury — liquid). *[Q144, Q146]*
- Metals ke paas metallic lustre hoti hai; ye good conductors of heat and electricity hote hain kyunki inme bahut se loosely bound electrons hote hain. *[Q144, Q146, Q148, Q224]*
- Metals ductile (wire ban sakte hain) aur malleable (sheet ban sakti hai) hote hain. *[Q148, Q150]*
- Metals ka boiling point aur melting point generally high hota hai. *[Q150]*
- Metals ke oxide aur hydroxides basic hote hain. *[Q145, Q149]*
- Metals electropositive hote hain kyunki ye electrons lose karke cations banate hain. *[Q157]*
- Element = wo matter jise chemical reactions se simpler substances mein todaa nahi ja sakta. *[Q197]*

2. Non-metals: Properties & General Characteristics

- Non-metals ke outermost shell mein 4–7 valence electrons hote hain (kuch options mein 5,6,7,8 bhi bataya jaata hai); ye electrons accept karte hain aur negative ions (anions) banate hain. *[Q143, Q145, Q147, Q271]*
- Non-metals gases ya brittle solids hote hain room temperature par (exception: Bromine — sirf non-metal jo liquid state mein hoti hai). *[Q144, Q146, Q177]*
- Non-metals non-lustrous hote hain (except Iodine — lustrous). *[Q148, Q152]*
- Non-metals poor conductors hote hain (except Graphite — good conductor). *[Q148, Q152, Q167, Q182]*

- Non-metals neither ductile nor malleable hote hain. *[Q150]*
- Non-metals ke oxide aur hydroxides acidic hote hain. *[Q145, Q149]*
- Non-metals se hydrogen gas ban sakti hai water ke saath reaction karke (jaise $C + H_2O \rightarrow CO + H_2$). *[Q145]*
- Non-metals solid, liquid, ya gas — teenon states mein exist kar sakte hain; sirf solid state mein hona non-metal ki physical property nahi hai. *[Q259]*
- Chlorine ek non-metal ka example hai. *[Q261]*

3. Reaction of Metals with Water

- Sodium (Na) water ke saath vigorously react karta hai: $2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$; exothermic reaction. *[Q142, Q154, Q222]*
- Potassium (K) sodium se bhi zyada vigorously react karta hai cold water ke saath; hydrogen gas burn ho jaati hai lilac flame ke saath. *[Q154, Q156, Q269]*
- Sodium aur Potassium dono open air mein rakhne par turant fire pakad lete hain — isiliye inhe kerosene oil ke andar store karte hain. *[Q245, Q257]*
- Calcium (Ca) water ke saath slowly react karta hai: $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2\uparrow$; calcium hydroxide (lime water) milti hai. *[Q154, Q156]*
- Magnesium (Mg) steam ke saath react karta hai; cold water se slowly. *[Q156]*
- Iron (Fe) steam ke saath react karta hai: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$. *[Q156]*
- Zinc (Zn) steam ke saath react karta hai: $Zn + H_2O \rightarrow ZnO + H_2$. *[Q156]*
- Aluminium bhi steam ke saath react karke metal oxide aur hydrogen deta hai. *[Q201]*
- Copper (Cu), Silver (Ag), Gold (Au) water se react nahi karte. *[Q156]*
- Reactivity order with water: K > Na > Ca > Mg > Zn > Fe > Cu. *[Q154, Q156]*
- ⚠️ Aluminium par ek thin protective oxide layer (Al_2O_3) ban jaati hai jo isse cold ya hot water ke saath directly react karne se roki hai — isliye "Al cold/hot water se react nahi karta" bhi sahi hai, jabki steam ke saath react karta hai. *[Q267]*

4. Reaction of Metals with Acids

- Reactive metals (Zn, Fe, Mg, Al) dilute HCl ya H_2SO_4 ke saath react karke hydrogen gas dete hain. *[Q158, Q160, Q162]*
- $Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$. *[Q158]*
- $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2\uparrow$. *[Q160]*
- $Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$ (Iron(II) chloride). *[Q162]*
- Copper (Cu) dilute HCl se react nahi karta kyunki ye hydrogen se kam reactive hai. *[Q162]*
- Copper concentrated H_2SO_4 se react karta hai: $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$. *[Q162]*
- Reactivity order with acids: Mg > Al > Zn > Fe > Cu. *[Q160, Q162]*

5. Reaction of Metals with Oxygen

- Metals oxygen ke saath react karke metal oxides banate hain. *[Q164, Q166, Q168]*
- Sodium oxygen ke saath jaldi react karta hai: $4Na + O_2 \rightarrow 2Na_2O$. *[Q164]*
- Magnesium air mein burn hota hai: $2Mg + O_2 \rightarrow 2MgO$; bright white flame. *[Q164, Q166]*
- Iron moist air mein rust karta hai: $4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$. *[Q166, Q168]*

- Copper heat karne par black copper oxide banata hai: $2Cu + O_2 \rightarrow 2CuO$. *[Q168]*
- Copper hawa mein slowly oxidize hoke green colour ki coating banata hai (copper corrosion). *[Q159]*
- Gold aur Silver oxygen se react nahi karte; inert hain. *[Q168]*
- Metal oxides generally basic hote hain; kuch amphoteric (Al_2O_3, ZnO). *[Q166]*
- Kuch metals (jaise alloys) high temperature par easily oxidize/burn nahi hote — pure metals ki compare mein zyada stable hote hain. *[Q209]*

6. Amphoteric Oxides & Hydroxides

- Amphoteric oxides both acid aur base ke saath react karte hain. *[Q170, Q172, Q174]*
- Examples: Al_2O_3, ZnO, PbO, SnO. *[Q170, Q172]*
- $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ (base ki tarah). *[Q170]*
- $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ (acid ki tarah). *[Q170]*
- $ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$. *[Q172]*
- $ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$. *[Q172]*
- Aluminium hydroxide $Al(OH)_3$ bhi amphoteric hai. *[Q174]*

7. Displacement Reactions & Reactivity Series

- Zyada reactive metal kam reactive metal ke salt solution se usko displace kar sakta hai — Metal A, Metal B ko displace kare to A, B se zyada reactive hai. *[Q176, Q178, Q180, Q263]*
- Complete reactivity series (high to low): K > Na > Ca > Mg > Al > Zn > Fe > Ni > Sn > Pb > H > Cu > Hg > Ag > Au > Pt. *[Q176, Q178]*
- $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ (Zn more reactive than Cu). *[Q176]*
- $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Iron displaces copper; blue solution green ho jaati hai). *[Q178]*
- $Cu + FeSO_4 \rightarrow$ No reaction (Cu less reactive than Fe). *[Q178]*
- $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$ (Copper displaces silver). *[Q180]*
- Zinc, Aluminium se kam reactive hai — reactivity series mein Al, Zn se upar hai. *[Q151]*
- Reactivity order examples: Ca > Al > Cu > Hg; K > Al > Fe > Pb > Au — high se low reactivity ke sequence yaad rakhna. *[Q153, Q161]*
- ⚠ Trap: Displacement reaction ke liye reactivity series compare karna zaroori hai; directly mat karo. *[Q176, Q178]*

8. Corrosion & Rusting Prevention

- Corrosion = metals ka atmosphere ke saath reaction karke gradually destroy hona. *[Q181, Q183, Q185]*
- Rusting = iron ka corrosion; moist air mein hota hai; rust ka formula: $Fe_2O_3 \cdot xH_2O$ (reddish-brown). *[Q181, Q183]*
- Rusting ke liye oxygen + water donon zaroori hain. *[Q183, Q185]*
- Prevention ke tarike: Painting (gates, bridges); Oiling/greasing (machinery); Galvanization — iron par zinc ki coating, sacrificial protection; Electroplating — chromium/nickel coating; Alloying — stainless steel (Fe + Cr + Ni + C). *[Q185, Q187, Q189]*
- Galvanized iron = iron coated with zinc; zinc pehle oxidize hota hai (sacrificial anode). *[Q187, Q189, Q213]*

9. Alloys: Composition & Uses

- Alloy = do ya zyada metals (ya metal + non-metal) ka mixture; properties pure metals se better hoti hain. *[Q190, Q192, Q194]*
- Brass = Cu (60–80%) + Zn (20–40%); utensils, decorative items, musical instruments. *[Q190, Q192, Q231]*
- Bronze = Cu (88%) + Sn (12%); statues, medals, bearings, bells. *[Q190, Q194]*
- Stainless Steel = Fe + Cr + Ni + C; utensils, surgical instruments, cutlery (rust-resistant). *[Q196, Q198, Q268, Q270]*
- Duralumin = Al + Cu + Mg + Mn; aircraft parts, lightweight structures. *[Q196, Q200, Q268, Q270]*
- Solder = Pb + Sn; electrical joints, plumbing (low melting point). *[Q198, Q200, Q211, Q233, Q272, Q274]*
- Amalgam = Hg + any other metal; dental fillings (Ag-Hg). *[Q202, Q247, Q272]*
- Gunmetal = Cu + Sn + Zn; bearings, valves. *[Q202]*
- Bell metal = Cu + Sn; bells, gongs (harder than bronze). *[Q204]*
- German silver / Nickel silver = Cu + Zn + Ni; decorative items, utensils. *[Q204]*
- Magnalium = Al + Mg; aircraft parts, lightweight, scientific instruments. *[Q206, Q270, Q272]*
- Constantan = Cu + Ni alloy; resistance wires mein use hota hai (resistance temperature ke saath almost constant rehta hai). *[Q205]*
- Nichrome = Ni, Cr, Mn aur Fe ka alloy; high resistance aur heat-resistant, electrical heating elements (heater, iron) mein use. *[Q237]*
- Steel ko hard aur strong banane ke liye Carbon add kiya jaata hai; Manganese bhi steel ko harden karne ke liye use hota hai. *[Q173, Q273]*

10. Extraction of Metals: General Principles

- Metals ores se extract kiye jaate hain; ore = metal-containing mineral + impurities (gangue). *[Q208, Q210, Q212]*
- Calcination = ore ko strongly heat karna absence of air mein; volatile impurities nikalte hain; carbonate ore → oxide. Example: $ZnCO_3 \rightarrow ZnO + CO_2$. *[Q210, Q212]*
- Roasting = ore ko heat karna presence of air mein; sulphide ore → oxide. Example: $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$. *[Q210, Q212]*
- Smelting = roasted/calcined ore ko reducing agent (C, CO) ke saath heat karna to get metal. Example: $ZnO + C \rightarrow Zn + CO$. *[Q212]*
- Flux = gangue ko slag mein convert karne ke liye substance add karte hain. Acidic gangue (SiO_2) ke liye basic flux (CaO, limestone); Basic gangue ke liye acidic flux (SiO_2). *[Q214]*
- Monazite Thorium ka important ore hai. *[Q171]*
- Limestone ek non-metallic mineral ka example hai. *[Q165]*

11. Extraction of Iron (Blast Furnace)

- Blast furnace mein iron extract hota hai haematite ore (Fe_2O_3) se. *[Q216, Q218, Q220]*
- Charge: iron ore + coke (C) + limestone ($CaCO_3$). *[Q216]*
- Reactions: Coke burn: $C + O_2 \rightarrow CO_2$; $CO_2 + C \rightarrow 2CO$ (reducing agent). Reduction: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$. Flux action: $CaCO_3 \rightarrow CaO + CO_2$; $CaO + SiO_2 \rightarrow CaSiO_3$ (slag). *[Q218]*
- Molten iron furnace ke bottom par collect hota hai; slag upper layer par (lighter). *[Q220]*
- Pig iron = impure iron; cast iron, wrought iron aur steel isse further process karke milti hai. *[Q220]*

12. Extraction of Aluminium (Hall-Héroult Process)

- Aluminium extract hota hai bauxite ore ($Al_2O_3 \cdot 2H_2O$) se. *[Q208, Q210]*
- Bauxite se pehle Al_2O_3 (alumina) purify karte hain Bayer's process se. *[Q210]*
- Electrolysis of molten alumina: Al_2O_3 dissolved in molten cryolite (Na_3AlF_6). Cathode: $Al^{3+} + 3e^- \rightarrow Al$; Anode: $2O^{2-} \rightarrow O_2 + 4e^-$; oxygen carbon anode se react karke CO_2 banata hai. *[Q208]*
- Cryolite add karne se melting point kam hota hai aur conductivity badhti hai. *[Q208]*
- Aluminium sirf electrolysis se hi extract kiya ja sakta hai — kyunki ye highly reactive metal hai. *[Q217, Q219]*
- Anodising process mein Aluminium ki surface par protective oxide layer banane ke liye dilute sulphuric acid electrolyte ke roop mein use hota hai. *[Q169]*

13. Extraction of Copper

- Copper extract hoti hai copper pyrites ($CuFeS_2$) se. *[Q214, Q216]*
- Roasting: $2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$.
- Bessemerization: $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$; phir $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$. *[Q214]*
- Blister copper = 98–99% pure; electrolytic refining se 99.9% pure milti hai. *[Q214]*

14. Metalloids & Their Properties

- Metalloids = metals aur non-metals ke beech ki properties rakhte hain; semiconductors hote hain. *[Q221, Q223, Q225]*
- Common metalloids: Boron (B), Silicon (Si), Germanium (Ge), Arsenic (As), Antimony (Sb), Tellurium (Te). *[Q221, Q223, Q155, Q175, Q195]*
- Silicon aur Germanium electronics mein semiconductors ke roop mein extensively use hote hain. *[Q225, Q227]*
- Metalloids ka lustre metallic hota hai lekin brittle hote hain; oxides amphoteric hote hain. *[Q223, Q225]*
- Alkali metals mein Lithium sabse chhota (smallest) hota hai size mein. *[Q191]*
- Silicon photovoltaic solar cells banane mein predominantly use hota hai. *[Q163]*
- Atomic number 14 wala element (Silicon) hard hota hai aur acidic oxide + covalent halide banata hai — isiliye Metalloid category mein aata hai. *[Q199]*

15. Non-metal Allotropes & Important Compounds

- Carbon ke allotropes: Diamond, Graphite, Fullerene, Graphene. *[Q226, Q228]*
- Diamond = hardest natural substance; sp^3 hybridization; bad conductor.
- Graphite = soft, slippery; sp^2 hybridization; good conductor (delocalized electrons); lubricant aur pencil lead mein use. *[Q228]*
- Sulphur ke allotropes: Rhombic (α-sulphur), Monoclinic (β-sulphur); Sulphur molecule polyatomic hoti hai (S_8). *[Q230, Q232, Q179]*
- Phosphorus ke allotropes: White (P_4), Red, Black. *[Q230, Q232]*
- White phosphorus = poisonous, glows in dark (chemiluminescence), stored under water. *[Q232]*
- Red phosphorus = non-poisonous, stable, matchbox striker mein use. *[Q232]*
- Nitrogen = inert diatomic gas ($N \equiv N$ triple bond strong); Haber process mein ammonia banane ke liye use. *[Q234, Q236]*
- Fluorine bhi ek diatomic non-metal hai. *[Q188]*

- Chlorine = greenish-yellow poisonous gas; water purification, bleaching powder ($CaOCl_2$) banane mein. *[Q234, Q236]*

16. Bleaching Powder, Baking Soda, Washing Soda & Plaster of Paris

- Bleaching powder = $CaOCl_2$ (Calcium oxychloride); chlorine ka smell; bleaching aur disinfecting mein use. *[Q238, Q240, Q242]*
- Manufacturing: $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$ (Back process). *[Q240]*
- Baking soda = $NaHCO_3$ (Sodium bicarbonate); mild base; baking, antacid, fire extinguisher mein. *[Q238, Q242, Q244]*
- Washing soda = $Na_2CO_3 \cdot 10H_2O$ (Sodium carbonate decahydrate); laundry, glass manufacturing, water softening mein. *[Q238, Q244, Q246]*
- Baking soda se washing soda banane ka reaction: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ (heat karke). *[Q244]*
- Plaster of Paris = $CaSO_4 \cdot \frac{1}{2}H_2O$; surgery mein fracture set karne ke liye; gypsum se banata hai ($CaSO_4 \cdot 2H_2O$ heat karke). *[Q246, Q248]*

17. Metals in Biological Systems

- Iron (Fe) = haemoglobin mein hota hai; oxygen transport blood mein. *[Q250, Q252]*
- Calcium (Ca) = bones aur teeth mein (as calcium phosphate); muscle contraction, nerve signaling; water se denser hota hai. *[Q250, Q252, Q193]*
- Magnesium (Mg) = chlorophyll ka central atom; photosynthesis ke liye zaroori. *[Q252, Q254]*
- Zinc (Zn) = enzymes (carbonic anhydrase, alcohol dehydrogenase) mein cofactor; immunity. *[Q254, Q256]*
- Copper (Cu) = haemocyanin (some animals mein oxygen carrier); enzymes mein; electrical wires banane mein bhi use hota hai (good conductor). *[Q256, Q243]*
- Iodine (I) = thyroid hormone (thyroxine) banane ke liye; goitre disease iodine deficiency se hoti hai; ye samundra mein abundantly milta hai. *[Q258, Q260, Q184]*
- Manganese ek trace element hai jo body mein thodi matra mein zaroori hota hai. *[Q186]*
- Sodium (Na) aur Potassium (K) = nerve impulse transmission; Na-K pump. *[Q258, Q260]*
- Fluorine (F) = teeth enamel strengthen karne ke liye; fluoride toothpaste mein. *[Q260]*

18. Acid Rain & Environmental Chemistry

- Acid rain = $pH < 5.6$; H_2SO_4 aur HNO_3 se banta hai. *[Q262, Q264, Q266]*
- Causes: SO_2 (fossil fuel burning) aur NO_x (vehicles) atmosphere mein jaake water ke saath acid banate hain. *[Q262, Q264]*
- Effects: marble/statues corrode ($CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$), aquatic life damage, soil fertility reduce. *[Q266]*
- Taj Mahal ka yellowing acid rain ki wajah se hota hai. *[Q266]*

19. Physical Property Records: Melting Point, Conductivity, Malleability & Ductility

- Metals mein sabse zyada malleable Gold (Au) hai; Silver (Ag) sabse zyada malleable metals mein bhi mention hota hai depending on comparison set — dono soft aur highly malleable metals hain. *[Q251, Q253, Q265]*
- Sabse zyada ductile metal Gold (Au) hai — sabse patla tar bana sakte hain. *[Q255]*

- Tungsten (W) ka melting point aur boiling point sabse zyada hota hai sab metals mein — isliye bulb ka filament Tungsten se banaya jaata hai. *[Q215, Q249]*
- Gallium (Ga) ka melting point bahut kam hota hai (near room temperature). *[Q239]*
- Silver compounds (jaise silver bromide) black and white photography mein use hote hain. *[Q241]*
- Electrical resistivity order: Hg > Ni > W > Ag (Mercury sabse zyada resistive, Silver sabse kam). *[Q203]*
- Lead (Pb) aur Mercury (Hg) heat ke comparatively poor conductor hote hain. *[Q235]*

20. ⚠ Traps / Key Differences Summary (Q142–Q274)

- ⚠ Trap: Mercury metal hai lekin liquid hai room temperature par — exception yaad rakhna. *[Q144, Q146, Q207]*
- ⚠ Trap: Graphite non-metal hai lekin good conductor hai — diamond non-conductor hai donon carbon ke allotropes hain. *[Q148, Q152, Q228, Q182]*
- ⚠ Trap: Iodine non-metal hai lekin lustrous hai — exception. *[Q152]*
- ⚠ Trap: Bromine ek hi non-metal hai jo liquid hai room temperature par. *[Q144, Q177]*
- ⚠ Trap: Sodium water ke saath H_2 gas deta hai, O_2 nahi; reaction exothermic hai. *[Q142]*
- ⚠ Trap: Copper dilute HCl se react nahi karta; concentrated H_2SO_4 se react karta hai. *[Q162]*
- ⚠ Trap: Iron dilute acid se $FeCl_2$ (Iron II) banata hai, $FeCl_3$ nahi. *[Q162]*
- ⚠ Trap: Al_2O_3 aur ZnO amphoteric hain — sirf basic nahi; acid + base donon se react karte hain. *[Q170, Q172]*
- ⚠ Trap: Displacement reaction mein zyada reactive metal salt solution mein displace karta hai; solid metal se nahi. *[Q176, Q178]*
- ⚠ Trap: Calcination = air absent mein heat; Roasting = air present mein heat — donon alag hain. *[Q210, Q212]*
- ⚠ Trap: Calcination carbonate ore ke liye; Roasting sulphide ore ke liye — ore type ke hisaab se choose karte hain. *[Q210, Q212]*
- ⚠ Trap: Flux gangue ko slag mein convert karta hai; slag lighter hoti hai aur upper layer par aati hai. *[Q214]*
- ⚠ Trap: Blast furnace mein coke reducing agent hai; CO_2 phir C ke saath react karke CO banata hai. *[Q218]*
- ⚠ Trap: Stainless steel mein chromium + nickel add karte hain rust resistance ke liye; sirf carbon nahi. *[Q196, Q198]*
- ⚠ Trap: Brass = Cu + Zn; Bronze = Cu + Sn — confuse mat hona. *[Q190, Q192, Q194]*
- ⚠ Trap: Solder ka melting point constituent metals (Pb, Sn) se kam hota hai — isliye joints mein use hota hai. *[Q198, Q200, Q272, Q274]*
- ⚠ Trap: Amalgam = mercury + any metal; dental filling mein silver amalgam use hota hai. *[Q202, Q272]*
- ⚠ Trap: White phosphorus poisonous aur water mein store karte hain; Red phosphorus stable aur matchbox mein. *[Q232]*
- ⚠ Trap: Bleaching powder = $CaOCl_2$; Baking soda = $NaHCO_3$; Washing soda = $Na_2CO_3 \cdot 10H_2O$ — teenon alag formulas hain. *[Q238, Q240, Q242, Q244, Q246]*
- ⚠ Trap: Plaster of Paris = $CaSO_4 \cdot \frac{1}{2}H_2O$; Gypsum = $CaSO_4 \cdot 2H_2O$ — $2H_2O$ vs $\frac{1}{2}H_2O$ difference hai. *[Q248]*
- ⚠ Trap: Chlorophyll mein Magnesium (Mg) central atom hai, Iron nahi — haemoglobin mein Fe hota hai. *[Q254]*
- ⚠ Trap: Goitre Iodine deficiency se hota hai; Fluoride deficiency se nahi — dental caries Fluoride se prevent hoti hai. *[Q258, Q260]*
- ⚠ Trap: Acid rain ka pH < 5.6 hota hai; normal rain ka pH ≈ 5.6 (CO_2 ki wajah se slightly acidic). *[Q262, Q264]*
- ⚠ Trap: Acid rain mein H_2SO_4 aur HNO_3 donon hote hain; sirf H_2SO_4 nahi. *[Q266]*
- ⚠ Trap: Aluminium cold/hot water se react nahi karta (protective oxide layer ki wajah se) lekin steam ke saath react karta hai — subtle difference. *[Q201, Q267]*
- ⚠ Trap: Sodium aur Potassium kerosene oil ke andar store karte hain kyunki ye open air mein turant fire pakad lete hain. *[Q245, Q257, Q269]*
- ⚠ Trap: Tungsten sabse zyada melting/boiling point wala metal hai — filament banane ke liye isiliye use hota

hai. *[Q215, Q249]*

- ⚠️ Trap: Gold sabse zyada malleable aur ductile metal hai — Silver bhi highly malleable hota hai, confuse na karein. *[Q251, Q253, Q255, Q265]*